NIMITI SHARMA

AI/ML Engineer | Deep Learning, Explainable AI & Generative AI | Applied Researcher

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PROFILE SUMMARY

B. Tech graduate in Artificial Intelligence & Machine Learning with hands-on experience across the full AI/ML lifecycle — data engineering, model development (CNN, GANs, Transformers), Explainable AI, and deployment as production-ready applications. Published across a Q1-indexed journal, a Scopus-indexed volume, and an international conference, with an ongoing cross institutional research collaboration (Umeå University, Sweden). Combines strong technical depth with proven leadership — as a hackathon team lead, conference reviewer, and student committee secretary — and is looking to bring that combination of applied AI expertise and ownership to a technology-enabled consulting environment.

CORE SKILLS

Machine Learning & Deep Learning: TensorFlow, Keras, PyTorch, Scikit-learn, CNN, DCNN

Generative AI & Synthetic Data: GANs (DCGAN, CGAN, CycleGAN), Agentic AI, LLM/RAG systems

Explainable AI (XAI): Grad-CAM, SHAP, LIME

NLP: NLTK, Transformers, BERT variants

Computer Vision: OpenCV, YOLO

Programming & Query Languages: Python, SQL, C

Data Analysis & Visualization: Pandas, NumPy, SciPy, Power BI, Tableau, Matplotlib, Seaborn, Plotly

Databases: MySQL, MongoDB

Deployment & Web: Flask, Streamlit, HTML, CSS, JavaScript

EXPERIENCE

AI/ML Project Intern — Attrix Technologies (Remote) *Dec 2024 – Mar 2025*

- Built and evaluated machine learning models end-to-end, from data collection and preprocessing to model validation, in a fully remote, self-directed environment.
- Delivered clear technical summaries to the project team, translating model outputs into decisions.

KEY AI/ML PROJECTS

Smart MindAlert — Brain Stroke CT Image Classification using Deep Learning & Explainable AI *Mar 2025*

- Led development of an interactive healthcare AI application benchmarking CNN, Xception, InceptionV3, MobileNetV2, and ResNet, reaching a peak accuracy of 97.21%.
- Integrated Explainable AI (Grad-CAM, SHAP, LIME) to make model predictions clinically interpretable, an AI chatbot, automated email alerts, and live video-consultation features to support early diagnosis.

Air Quality Index (AQI) Prediction — Machine Learning, IoT *Nov 2022 – Dec 2024*

- Built an end-to-end pipeline that ingested live IoT sensor data, processed it, and applied ML models to predict AQI, extracting actionable environmental insights.

Social Distancing Monitoring — Deep Learning, OpenCV *Nov 2023*

- Developed a YOLOv3-based computer vision framework to detect and flag social-distance breaches in real time from overhead video feeds.

Home Safety with Gas Sensing Technology — IoT, OpenCV *Mar 2024*

- Built a Raspberry Pi and OpenCV-based fire/gas detection system to enable early hazard warnings for home safety.

Virtual Personal AI Assistant — NLP, AI *Mar 2024*

• Developed a Python-based virtual assistant capable of processing natural language commands to send messages/emails, fetch news and weather, and respond to varied user queries.

Plant Leaf Disease Detection — Deep Learning, Python *Mar 2024*

• Trained a CNN model for plant disease classification and deployed it as a real-time Streamlit web application.

ONGOING RESEARCH PROJECTS

- Medical Brain Tumor MRI Image Generation Using GAN Variants — in collaboration with a researcher from Umeå University, Sweden.
- Explainable AI & Deep Learning Framework for Multi-Class Plant Disease Detection across 38 crop categories.
- Agentic AI-Based Small Language Model (SLM) for Legal Support.
- Prompt Injection Analysis Using BERT Variants.

RESEARCH PUBLICATIONS

- "Review on Air Quality Index and Air Pollutant Concentration Prediction based on IoT and AI/ML Algorithms" — Indoor Air (Q1 Journal), 2026 (**Manuscript Under Review**).
- "Network Security and Data Privacy in 6G Communication: Japan's 2030 Vision for 6G – A Case Study" — Taylor & Francis (Scopus-Indexed), 1st Edition, 2025 (First Author).
- "BNS Mitra: RAG-Optimized LLM-Based AI-Powered Legal Virtual Assistant" — ICCSAI 2025 (International Conference).

EDUCATION

B. Tech, Computer Science Engineering — Specialization in AI & ML *2021 – 2025*

• Unitedworld Institute of Technology, Karnavati University — CGPA: 8.05/10

Higher Secondary School (Science) — Gujarat Secondary Education Board *2021*

LEADERSHIP & ACADEMIC SERVICE

- Reviewer, IEEE Contemporary Computing Innovations Conference (CCIC) 2026 — evaluated technical submissions for research quality and rigor.
- Secretary, USCI Events and Media Committee, Karnavati University (2 years) — coordinated cross-team logistics and communications for university-wide events.
- Smart India Hackathon — Led a team to 4th position among top 10 teams (Mar 2022).
- CodeUnnati Innovation Marathon (GTU) — Semi-Finalist Team (Mar 2024).

CERTIFICATIONS & COURSES

- Supervised Machine Learning Course — Coursera
- HTML, CSS, and JavaScript for Web Developers — Coursera
- Edunet SAP Foundation Advance Course — CodeUnnati / Edunet Foundation
- Introduction to Deep Learning and AI Accelerations — CloudxLab
- Introduction to Develops and its Application — CloudxLab
- Java Programming; Python Fundamentals for Beginners; Introduction to R — Great Learning